

WHEN IS THE SEX OF A FETUS FIXED?

Chromosomal sex is determined at the point of fertilization. Physical sexual differentiation occurs seven to 12 weeks after fertilization.

Physical differences

Differences between males and females begin with the sex chromosomes at the moment of conception: XX for females and XY for males. In the uterus, the release of testosterone from the mother during gestation “masculinizes” a male fetus, triggering the growth of structural sex differences in both the brain and body. As we grow and develop, these differences will arise in many different brain structures (see right). Cognitive and skill differences between the sexes are present from childhood. Adult male brains are 8 to 13 percent larger, on average, than adult female brains. In addition, adult male brains also tend to vary more, in volume and cortical thickness, than female brains.

Male and Female Brains

Scientists have found that male and female brains show distinct physical differences. However, it is not always clear how these variations affect our attitudes, activities, and responses to our environment. Differences may arise from the way a brain is used in life as well as from its physical form.



LARGER IN MALE BRAIN

Thalamus

This area, the “relay station” between the cortex and deeper brain structures, is larger in men than in women. The two sides of the thalamus are more likely to be connected in females, but the significance of this feature is not known.



LARGER IN FEMALE BRAIN

Corpus callosum

The corpus callosum, which links the brain’s left and right hemispheres, has been found to be larger in females. It has been associated with greater cognitive skills in females, possibly because brain functions are shared between hemispheres, but not in males.



LARGER IN MALE BRAIN

Hippocampus

Males have a larger anterior (front) hippocampus, which governs acquiring and encoding new spatio-visual information, while females have a larger posterior hippocampus, which governs retrieval of existing spatio-visual knowledge.

ALL HUMAN EMBRYOS START LIFE WITH FEMALE BRAINS—EXTRA HORMONES ARE NEEDED TO CREATE A MALE

